

WAVE OPTICS

Important formulas and concepts directly from the NCERT-

- According to corpuscular theory, if ray of light bends towards the normal then the speed of light would be greater in the second medium.
- This theory explained the laws of reflection and refraction.
- **Dutch scientist Huygens put forward ‘wave theory of light’**, it predicted that if light bends towards the normal then speed would be less in the second medium.

This is in contradiction to corpuscular theory.

❖ Huygens principle-

- **Wavefront-** locus of all points which oscillates in same phase.
- Types of wavefront-
 - 1) **Spherical wavefront-** if source is point source.
 - 2) **Cylindrical wavefront-** if point is linear.
 - 3) **Planer wavefront-** if source is at large distance.
- According to this principle, each point of the wavefront is the source of a secondary disturbance and the wavelets emerging from these points spread in all directions with speed of wave.
- The wavelets are usually referred to as ‘secondary wavelets’.
- Common tangent on secondary wavelets forms a new position of wavefront.

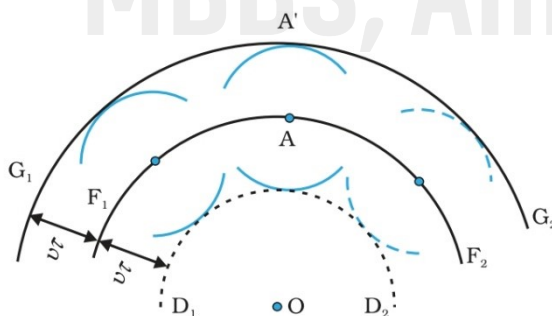
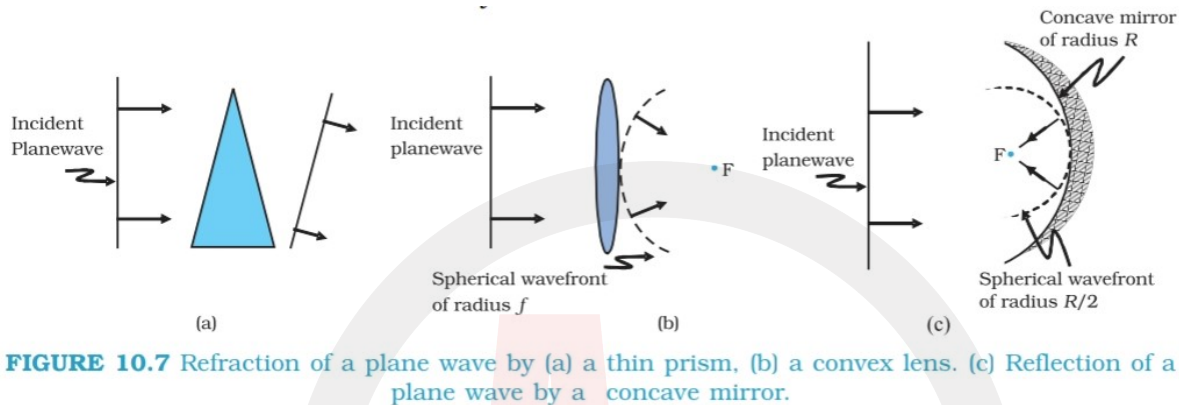


FIGURE 10.2 F_1F_2 represents the spherical wavefront (with O as centre) at $t = 0$. The envelope of the secondary wavelets emanating from F_1F_2 produces the forward moving wavefront G_1G_2 . The backwave D_1D_2 does not exist.

- Acc. to Huygens, amplitude of secondary wavelets are maximum in forward direction and zero in backward direction.
- **when a wave gets refracted into a denser medium, the wavelength and speed of propagation decrease but the frequency remains same.**



❖ Coherent and Incoherent addition of waves-

- “two sources are coherent if phase difference(ϕ) at any point doesn't change with time”.
- Interference is based on the superposition principle according to which the resultant displacement is the vector sum of the displacements produced by each of the waves.
- The amplitude of resultant displacement,

$$y_{\text{net}} = a \cos(\phi/2)$$

- Resultant intensity,

$$I_{\text{net}} = 4 I_0 \cos^2(\phi/2)$$

❖ Interference of light waves and Young's experiment-

- If two coherent waves are superimposed on each other when they are travelling in same direction then intensity of light is different at different points, this phenomenon is k/as ‘Interference’.

- Fringe width, $\beta = \lambda D/d$

- **Central maxima width = $\lambda D/d$**

➤ Interference is of 2 types-

CONSTRUCTIVE INTERFERENCE	DESTRUCTIVE INTERFERENCE
• phase diff, $\Delta\phi = n(2\pi) = 0, 2\pi, 4\pi, 6\pi$ etc. ($n = 0, 1, 2, 3, \dots$) • path diff, $\Delta x = n\lambda = \lambda, 2\lambda, 3\lambda$ etc. ($n = 1, 2, 3, \dots$) • resultant intensity, $I_{\text{net}} = 4I_0$ • resulting in bright fringe (maxima) position of bright fringe, $y = n\lambda D/d$ ($n = 0, 1, 2, 3, \dots$) here, D = distance of screen from source and d = distance b/w holes(sources)	• phase diff, $\Delta\phi = (2n+1)\pi = \pi, 3\pi, 5\pi$ etc. ($n = 0, 1, 2, 3, \dots$) • path diff, $\Delta x = (2n+1)\lambda/2 = \lambda/2, 3\lambda/2, 5\lambda/2$ etc. ($n = 0, 1, 2, 3, \dots$) • resultant intensity, $I_{\text{net}} = 0$ • resulting in dark fringe (minima) position of dark fringe, $y = (2n+1)\lambda D/2d$ ($n = 0, 1, 2, 3, \dots$)

❖ diffraction-

- light turns around corners and enter regions where we would expect a shadow. These effects are k/as diffraction.
- There is a central maxima at ($y=0$)
- Other secondary maxima at $y = (2n+1)\lambda D/a$ (a = width of single slit)
Where ($n = 1, 2, 3, \dots$)
- Position of minima, $y = n\lambda D/a$ ($n = 1, 2, 3, \dots$)
- Angular position of secondary maxima, $\theta = (2n+1)\lambda/a$
Angular position of minima, $\theta = n\lambda/a$
- Fringe width, $\beta = \lambda D/a$
- **Central maxima width = $2\lambda D/a$**

❖ Polarisation-

- Wave eqn, $y = a \sin(kx - \omega t)$
Here, $k = 2\pi/\lambda$ (λ = wavelength of wave)
- If plane of vibration of string is changed randomly in very short intervals of time, it is k/as '**unpolarised wave**'.
- If each point on string moves on a straight line or the string always remain confined to the x-y plane, the wave is referred as '**linearly polarised wave**' or '**plane polarised wave**'.
- **A polaroid transmits only one component. If unpolarised light pass through a polaroid, it converts into a plane polarised light.**
- **Malus Law,**

$$I = I_0 \cos^2\theta$$

Where, I_0 = intensity of the polarised light after passing through polaroid.

θ = angle b/w plane polarised light and polaroid.

Dr. Aman Tilak
MBBS, AIIMS Delhi

SUMMARY

1. Huygens' principle tells us that each point on a wavefront is a source of secondary waves, which add up to give the wavefront at a later time.
2. Huygens' construction tells us that the new wavefront is the forward envelope of the secondary waves. When the speed of light is independent of direction, the secondary waves are spherical. The rays are then perpendicular to both the wavefronts and the time of travel is the same measured along any ray. This principle leads to the well known laws of reflection and refraction.
3. The principle of superposition of waves applies whenever two or more sources of light illuminate the same point. When we consider the intensity of light due to these sources at the given point, there is an interference term in addition to the sum of the individual intensities. But this term is important only if it has a non-zero average, which occurs only if the sources have the same frequency and a stable phase difference.
4. Young's double slit of separation d gives equally spaced interference fringes.
5. A single slit of width a gives a diffraction pattern with a central maximum. The intensity falls to zero at angles of $\pm\frac{\lambda}{a}, \pm\frac{2\lambda}{a}$, etc., with successively weaker secondary maxima in between.
6. Natural light, e.g., from the sun is unpolarised. This means the electric vector takes all possible directions in the transverse plane, rapidly and randomly, during a measurement. A polaroid transmits only one component (parallel to a special axis). The resulting light is called linearly polarised or plane polarised. When this kind of light is viewed through a second polaroid whose axis turns through 2π , two maxima and minima of intensity are seen.

POINTS TO PONDER

1. Waves from a point source spread out in all directions, while light was seen to travel along narrow rays. It required the insight and experiment of Huygens, Young and Fresnel to understand how a wave theory could explain all aspects of the behaviour of light.
2. The crucial new feature of waves is interference of amplitudes from different sources which can be both constructive and destructive, as shown in Young's experiment.
3. Diffraction phenomena define the limits of ray optics. The limit of the ability of microscopes and telescopes to distinguish very close objects is set by the wavelength of light.
4. Most interference and diffraction effects exist even for longitudinal waves like sound in air. But polarisation phenomena are special to transverse waves like light waves.

Dr. Aman Tilak
MBBS, AIIMS Delhi